

What is NetBIOS Enumeration?

NetBIOS Enumeration is the process of gathering **network-related information** about a **Windows system** using the **NetBIOS (Network Basic Input/Output System)** protocol.

In Simple Words:

NetBIOS is like the "old-school" way Windows computers used to **find and talk to each other on a local network**.

NetBIOS Enumeration means extracting info like:

- Computer names
- Usernames
- Shared folders
- Network domains

NetBIOS Ports

Port Protocol Use

137	UDP	NetBIOS Name Service (NBNS)
138	UDP	NetBIOS Datagram Service
139	TCP	NetBIOS Session Service (SMB)

These are the **main ports** NetBIOS operates over.

What Can Be Enumerated?

Info You Can Collect

Example

 Computer Name	DESKTOP-USER1
 Domain/Workgroup Name	WORKGROUP, CORP.LOCAL
 Logged-in Usernames	admin, johndoe
 Shared Folders/Resources	\\target\public , \\target\C\$

✅ MAC Addresses

00:1A:2B:3C:4D:5E

Tools for NetBIOS Enumeration

Tool	Description
nbtscan	Scans network for NetBIOS info
nmblookup	Linux command to query NetBIOS names
Net View	Windows command to list shared resources
NBTStat	Shows NetBIOS names, MAC, and sessions
enum4linux	Uses NetBIOS along with SMB to gather info

Examples

1. nbtscan (Linux)

```
nbtscan 192.168.1.0/24
```

➡ Scans the whole subnet for NetBIOS names, MAC addresses, and users.

2. nmblookup (Linux)

```
nmblookup -A 192.168.1.10
```

➡ Queries a single target for NetBIOS names and shares.

3. NBTStat (Windows)

```
nbtstat -A 192.168.1.10
```

➡ Displays remote machine's NetBIOS table and MAC address.

Why Attackers Use NetBIOS Enumeration

 Goal	Benefit
✅ Find computer names	Useful in crafting phishing or social

✓ Discover shared folders	engineering
✓ List users/groups	May contain sensitive info
✓ Identify domain/workgroup	For password spraying or login attempts
	Useful in AD and network mapping

How to Prevent NetBIOS Enumeration

Action	Description
 Disable NetBIOS over TCP/IP	It's outdated and often unnecessary
 Use firewall rules	Block ports 137, 138, 139 externally
 Restrict share permissions	Disable guest/anonymous access
 Monitor logs	Detect NetBIOS or SMB scans

✓ Summary

Term	Meaning
NetBIOS	Protocol used for name resolution in older Windows systems
Enumeration	Gathering system names, users, shares, etc.
Tools	nbtscan, nbtstat, enum4linux, etc.
Usage	Common in footprinting and internal attacks